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ORIGINAL RESEARCH

Novel Insights into the Underlying Neurobiology of Addiction

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Abstract :

In recent years, the rise of opioid use disorder has garnered significant attention. Despite opioid replacement therapy being a commonly used method for treating addiction, it has been noted to be ineffective in many cases. Dr. Angelika Wagner's clinical trial provides hope for those battling addiction, particularly in cases of opioid use disorder. According to the trial's findings, trauma seems to be the root cause of pre-addictive behavior in many cases. Furthermore, the examination of the brain's dopamine receptors has offered a totally new approach to non-invasive treatments. The trial's successful short-circuiting and subsequent effect on dopamine receptors through electro-acupuncture provides a promising alternative to conventional methods. These results signify the potential for non-invasive methods to effectively treat addiction, emphasizing the importance of continued research and advancements in this field.

Keywords :

Psychology, Neurobiology, Neuropharmacology of Addiction.

Introduction:

Case Study of a Country in Trauma, Japan Analysis of the Psychology, Neurobiology and Neuropharmacology of Addictions

Since 1995, with the beginning of the third epidemic of methamphetamine abuse, drug abuse or dependence in Japan has changed markedly. “During WWII, meth use was encouraged by the Japanese state to enhance the productivity of factory workers and to instill national duty to fightin battles,” Brewstersaid. “Following the Japanese defeat, the use of the drug spiraled into an epidemic. The act of choosing illegal drugs was deemed to be innately selfish, socially and morally polluting.”

What other incidents in countries involved in civil war, political distress or underperforming socioeconomic issues minimize human excellence due to drug use? In California alone, the cost of housing and maintenance of a drug addict is $107,000 per year.

Subject Line:

Addiction, Brain Reward Cascade, clinical trial, innovation, hertz frequency, recidivism, sober living, meth, cocaine, heroin, substance abuse, alcohol, Japan, cultural norms, education, monoterpenes, sesquiterpenes.

Background: Theory of Root Cause

Precepts in the Psychology of Non- Relapse Addiction:

Is non-relapse in addiction possible?

If so, is permanent non- drug, non- pharmaceutical addiction recidivism achievable?

The initial clinical trial took place from 2013 -2023, with 95 incarcerated people from a sober living facility in Los Angeles, California participating. Three different addiction groups were subsequently studied over a period of 10 years. Based on an incarcerated population, no tracking, no controls: all participants exhibited socio-occupational impairments, mental and psychological debilitation due to addiction drug use with comorbidity of brain and organ dysfunctions (Von Fluegge, Balance Atlanta: Addiction a Unique Chiropractor's Pursuit of the Course).Based on a controlled incarcerated population, all participants experienced social, emotional, chemical or physical trauma prior to engaging in addiction behavior (Van Emmerik-van Oortmerssen, et al., 2014). Based on an adolescent suicidal population, all participants were medicated through legal prescriptions by psychiatrists due to emotional trauma (Matthews, Dwyer,& Snoek, 2017). From 2017-2021 further research with adolescents of suicidal ideation in Toronto, took place in private practice, in Canada.

The precept of these studies originated in a sober living community based on individual study of the socio-economic examination of the psychological, neurobiological and pharmacological aspects of inmate addiction. Due to a subsequent macro level hypothesis, the chemical dependency of individuals was later linked to the trauma and chemical dependency of their resident country. A preliminary abstract of Japan as a traumatized country examines emerging trends of drug manufacture co-related to the prevalent populous sector of use. Participants exhibited mental debilitation with multiple drug addictions, years of isolation and incarceration, movement disorders, vision disorders, inability to work, and deep depression. Of the 95 participants, 33/35 participants who remained in the trial, have remained sober since 2013.

In a separate clinical trial in Toronto, Canada with 6 adolescent girls exhibiting suicidal ideation disorder social trauma requiring prescription SSRI use with suicidal ideation was an outcome.

The adolescent study ran from 2017-2021; parallel protocols and parallel results occurred. These similarities were not coincidental, but led to cross sector conclusions in divergent mental health sectors bearing the same root cause.

If the root cause of trauma is the same in all populations, it becomes a waste of time for researchers to spend endless decades seeking different outcomes for different populations based on an incorrect theory of cause.

The predisposition of drug use due to trauma was conclusive in all populations. This led to the examination of the neurobiology and neuropharmacology affecting the current toxic treatments, followed by the examination of trauma in specific countries.

In the United States alone, it is estimated that over two million people per year suffer from opioid use disorder resulting from illegal prescription use (Chung, 2018) (Nutt, Lingford-Hughes, Erritzoe, & Stokes, 2015) (Volkow & Blanco, 2021) (Britch & Walsh, 2022). It is also estimated that the overall cost of these legally and illegally prescribed opioids is an ongoing crisis exceeding one trillion dollars globally (Rehm & Shield, 2019) (Coussens, et al., 2019) (Gardner, McGrath, Dowling, & Bai, 2022) (Manchikanti, et al., 2018) (Manchikanti, et al., 2022). Although Opioid replacement therapy (Manchikanti, et al., 2018) (Manchikanti, et al., 2022) (Busserolles, et al., 2020) (Coluzzi, et al., 2022) is currently the only common treatment therapy for opioid addiction, the method was found to be ineffective, causing escalating opioid use and comorbidity of the liver, stomach, and colon over the long term (Chung, 2018) (Nutt, Lingford-Hughes, Erritzoe, & Stokes, The dopamine theory of addiction: 40 years of highs and lows, 2015) (Rehm & Shield, 2019) (Volkow & Blanco, 2021) Even after patients repeatedly relapsed, prescription use of the same drugs continued (Hayashi, et al., 2021) (Chisholm-Burns, Spivey, Sherwin, Wheeler, & Hohmeier, 2019) (Gardner, McGrath, Dowling, & Bai, 2022) (Manchikanti, et al., 2018). Alternative therapies were not recommended due to scientific bias.

Concurrently in 2013-2017, suicides in veterans (Vargas- Medrano, et al., 2020), adolescents (Van Emmerik-van Oortmerssen, et al., 2014) (Stolberg, 2002) (Marshall, 2000) (Pelham, et al., 1999) (Vargas-Medrano, et al., Psychological and neurobiological aspects of suicide in adolescents: Current outlooks, 2020) and children spiked due to prolonged trauma and prolonged SSRI use (Vargas- Medrano, et al., Psychological and neurobiological aspects of suicide in adolescents: Current outlooks, 2020)

(Hetrick, Merry, McKenzie, Sindahl, & Proctor, 2007) (Comer & Cahill, 2019) (Becker & Correll, 2020) Research shows that over 50% of suicides in adolescents are caused by drug and/or alcohol addiction as early as 9-12 years of age, and escalating in adolescent years.

Veterans at Camp Pendleton were dying by suicide at a rate of 25 per week, as late as 2017. Adolescents in Orange County, California were dying at the rate of 20 per month due to drug overdose/suicide simultaneously. Parent Teacher associations denied saying, "We do not have a problem." (Sharma & Couture, 2014) (Sprich, Safren, Finkelstein, Remmert, & Hammerness, 2016) (Findling, 2008) (Konstenius, et al., 2017) (Van Emmerik-van Oortmerssen, et al., 2014). How long do we deny ovedose as suicide? A judge recently ruled that increased addiction to internet use in high functioning countries like Japan, India, China and USA is also causing "grave social harm," and adolescents are committing suicide due to social pressure and bullying (Kuss, Griffiths, Karila, & Billieux, 2014) (Jorgenson, Hsiao, & Yen, 2016) (Derevensky, Hayman, & Gilbeau, 2019) (Schou Andreassen, et al., 2016) (Kuss, Griffiths, Karila, & Billieux, 2014) (Cohen, 2009).

What will it take for global leaders in science and technology to be held accountable for the increasing numbers of deaths caused by prescription addiction? (Chisholm-Burns, Spivey, Sherwin, Wheeler, & Hohmeier, 2019) (Gardner, McGrath, Dowling, & Bai, 2022) (Manchikanti, et al., 2018) (Manchikanti, et al., 2022) (Busserolles, et al., 2020).

What will it take to stop feeding Benzodiazepines to babies like "skittles" candy citing ADHD and depression? The recent Ritalin clinical trials are serving tots from 3-5 years old. (Sprich, Safren, Finkelstein, Remmert, & Hammerness, 2016) (Findling, 2008) (Konstenius, et al., 2017) (Van Emmerik-van Oortmerssen, et al., 2014) (Stolberg, 2002) (Pelham, et al., 1999) (Busby, Hatkevich, McGuire, & King, 2020).

The practice of prescription medication of very young children's ADHD comprises 59% of benzodiazepine use in the world, occurring in populations from 6-16 years of age. Facts show the global incidents of suicide in children between the age of 10 and 16 is the third leading cause of death in children in 2023 (Busby, Hatkevich, McGuire, & King, 2020) (Vargas-Medrano, et al., Psychological and neurobiological aspects of suicide in adolescents: Current outlooks, 2020) (Lewis, et al., 2021) (Lunghi, et al., 2022) (Cousins & Goodyer, 2015) (Hetrick, Merry, McKenzie, Sindahl, & Proctor, Selective serotonin reuptake inhibitors (SSRIs) for depressive disorders in children and adolescents, 2012).

The USA leads the race consuming five times the entire world consumption in ADHD medications with 11% of children addicted. Canada is in 4th place with 8% of the population addicted. The addiction of populations in countries is linked to SSRI use in children.

THE PERCENTAGE of Children with Antidepressant USE Globally

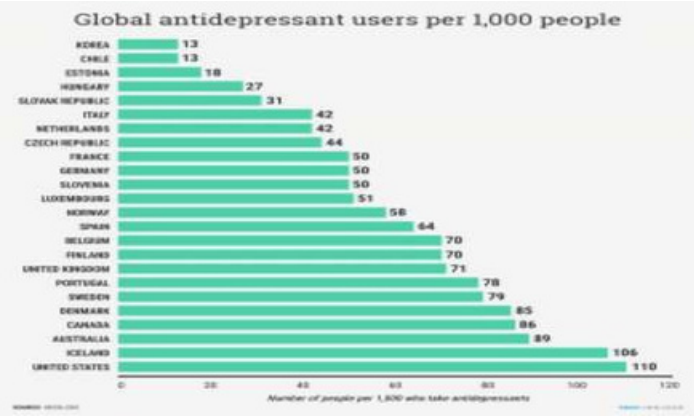

Figure 1. (Google Images)

Aims :

● To determine novel root causes of addiction in cross-cultural populations of mental health instability.

● To determine novel outcomes of non-pharmaceutical methods and materials to create permanent recidivism.

● To determine innovative new high-frequency points through auricular acupuncture to create "rewire" on the neuronal patterns.

● To determine advanced biology in plant medicines supporting populations in addiction and suicide recovery

● To educate the public and the global scientific community of the neurobiological and neuropharmacological toxicity effects of Meth, Cocaine, opioids, and synthetic cannabis on the brain and organs.

● To educate the public and global scientific community on the debilitation of adolescent neurobiology and brain capacity through prolonged use of SSRI and Ritalin use in children creating suicidal ideation.

● To create public awareness that nondrug addiction and suicidal ideation impairments can be permanently reversed in 90 days with a lifetime guarantee.

Methods and Materials (Primary Protocols of Care)

This pilot open-protocol study examined the possibility of achieving non-relapse recidivism in the addictions of meth, cocaine, opioids and alcohol through the following methods that were tailored to individual needs.

● Trauma collapse techniques including CBT, DBT, and Access Consciousness.

● These Trauma release methods were tailored to the needs of each participants

● Stim Pro plus, electronic auricular acupuncture stimulation 30 hertz device on Limbic, Parasympathetic, Hypothalamus, Thalamus, and Pituitary points, liver, kidney points. *Auricular Acupuncture use for 90-120 consecutive days, or until the heat readings on the Stim Pro device reach zero for a minimum of three days. Tailored to individual needs.

● Used 30 hertz (Hz) frequency Stim Pro-plus on cranial nerves 3, 4, 6 to improve sight of individuals, and 5, 9, 10 to improve vagal tone, digestion and heart coherence. *Used 40-hertz biophotonic modulation device (Vielight/Toronto)

● Pure essential oils with frequencies of 160-660 Hz inhaled were used as inhalation substitutes. The exact formulations were tailored to the specific needs of the participation. *In every individual the heart, organs, and sensory lobes were affected by the co-morbidity of drug use. Unless these areas of care were addressed, the individual would not feel well enough to return to work or study.

Discussion

Topic # 1: The Psychology of Addiction:

1. Could the trauma of an event in childhood have caused the self medication, or use of the initial opioid, drug or alcohol in order to escape the memory of it?

2. If the awareness of this event, erasing or "lack of memory" of the event could be achieved, would the person still relapse into drug use to escape the memory and psychological precedent?

3. Once the trauma was "deleted" from the memory would d new cognitive behavior therapy be able to achieve recidivism?

4. Was deletion of the trauma from the trauma loop of the brain solely responsible to eliminate recidivism permanently?

5. Does the trauma of a specific country influence specific types of addiction to proliferate the population, including the behavior of drug dealers? These country traumas included wars, natural disasters, environmental disasters, political or governmental incapacities of care, socio-economic financial depletion due to the Covid pandemic.

Conclusion of Protocol #1: Eliminating the root cause of addiction; Childhood trauma. According to talk therapy specialists these traumas can be talked away after several years, sometimes decades, creating an inner journey of enlightenment as the "victim" self medicates in order to cope with the memories of their life. If the emotional impact could be neutralised would this person be able to function without the need to escape into drug use. Are there protocols that short circuit talk therapy and shorten the time for the trauma to be neutralised.

And, is there a method that could delete, or erase the trauma without the years of tak therapy recommended. Even after learning 9 different modalities of trauma therapy, the easiest for each person was Access Consciousness. The method of acupressure on the scalp points called Access Consciousness, not only reduced and elinminated the post traumatic disorder, but in most cases, the person could not remember the incident, or if remebered, rememberd it in a new context. Other conventional therapies such as CBT, and DBT reduced and eliminated the elicitation of post-traumatic disorder. (PTSD). Therefore, participants without the triggers of repeat trauma experienced a greater sense of ease, well-being and coping skills. Trauma collapse alone was not found to be the sole contributor of a permanent end to the recidivism of the disease.

Discussion #2: Neurobiology of Addiction with Disruption of the Neuronal Trauma Loop using Auriculotherapy.

This Sequence of the Treatment Protocol is evidence based on Brain Reward Cascade Theory with Disruption of the Neuronal Circuits :

The initial clinical trial involved 95 incarcerated participants from the Los Angeles jail with permission of the LA Sheriff's office from May-November in 2013. Although 95 participants began the clinical trial, only 35 remained for 90-120 days. Sixty participants did not complete a full 90 days of clinical trial. These people left after a few sessions and never came back and were not included in the final results.

Thirty-two male participants and three female participants ranging in age from 23-61 with chronic drug/alcohol abuse or opioid addiction requiring incarceration. All participants had non-penetrating brain trauma, co-morbidity of organ damage reported through survey.

Restrictions in employment due to lack of movement/somatosensory control caused by chemical brain poisoning. All participants had a history of 8-28 years incarcerations due to drug/alcohol abuse.

The participants had forty outpatient sessions (Monday, Wednesday, Friday, for 20 weeks. Participants reported non craving of abusive substances 10 days or less, improved cognitive ability, improved sleep, fewer post-traumatic stress disorder (PTSD) symptoms, improved mobility and sensory input.

Hypothesis Questions:

1) Was it possible for the neurobiology of addictive behavior be disrupted by electronic stimulation of the trauma axis in the Brain Reward Cascade?

2) What effect would high frequency stimulation (30 Hertz) of the cranial nerves restore sensory somatic regulation in the brain, restoring sensory organs of the participants

3) What effect would near red and infra red light therapy (40 Hertz) have to restore motor movement, somatosensory capacity, executive function and occipital lobe capacity after the chemical high?

Epidemiology of Care: Neurobiology of Addiction

Research done at Harvard University studies have shown abnormal frontal lobe activation in addiction patients. Two regions within the frontal lobes susceptible to damage during addiction are the prefrontal cortex and the anterior cingulate gyrus (Tafet & Nemeroff, 2020) (DeSabel, A, Kruse, Wolf, & Guenther, 2013).

In 1990, Dr. Jay Holder (13) discovered that addiction could be mitigated holistically through tonal placement of electro- acupuncture on the exterior of the ears. He discovered the limbic and parasympathetic systems could achieve balance through electronic stimulation.

Repeated application of these methods created short term brain balance, but not a total recidivism result Dr. Angelika Wagner was seeking. The prefrontal cortex is involved in maintaining, monitoring, and manipulating information in working memory and particularly in sustaining attention. The anterior cingulate gyrus has been implicated in divided attention, novelty detection, working memory, memory retrieval, Stroop tasks (inhibition), evaluative judgment, motivation, and performance (Krentzman, 2013) (Werner, Altshuler, Shaham, & Li, 2021) (Frankiensztajn, Elliott, & Koren, 2020) (DeSabel, A, Kruse, Wolf, & Guenther, 2013).

Clinical hypothesis for the trial was based on the study of the Brain Reward Cascade discovered by Dr. Kenneth Blum in 1995. Dr. Angelika Wagner studied with Dr. Kenneth Blum and Dr. Jay Holder from 1995-2013. (DeSabel, A, Kruse, Wolf, & Guenther, 2013) The team discovered that the functioning of brain dopamine receptors could be short circuited with amino acids nutrition. (Frankenstajn, Elliott, & Koren, 2020) and electro-acupuncture of the primary addiction points and the HPA axis (DeSabel, A, Kruse, Wolf, & Guenther, 2013). In 1995, the protocol of the short circuiting the trauma loop had not been addressed.

The Brain Reward Cascade is a complex system involved in the processing of pleasurable experiences and the reinforcement of behavior (Roberto, Kirson, & Khom, 2021) (Gilpin, Herman, & Roberto, 2022) (Wolf, 2016) (Antonelli, et al., 2021) (Blum, et al., 2022) (Febo, et al., 2017) (Roy, et al., 2021). It is primarily mediated by the release and interaction of various neurotransmitters within specific brain regions. One key component of the Cascade is the meso-limbic dopamine pathway (Blum, et al., 2022) (Febo, et al., 2017) (Febo, et al., 2017) (Gong, Fayette, Heinsbroek, & Ford, 2022) (Groman & Jentsch, 2012) (Chamberlain, et al., 2019) which plays a central role in motivation, reward, and addiction (Krentzman, 2013).

The limbic system is controlled by the emotions of the person escaping reality in his world (Roberto, Kirson, & Khom, 2021) (Gilpin, Herman, & Roberto, 2022) (Wolf, 2016). This creates the neverending, relentless trauma loop causing the addiction.

Therefore, the release of dopamine within the nucleus accumbens (NAc), a region in the ventral striatum, is a crucial event in the Brain Reward Cascade (Chung, 2018) (Nutt, Lingford-Hughes, Erritzoe, & Stokes, The dopamine theory of addiction: 40 years of highs and lows, 2015).

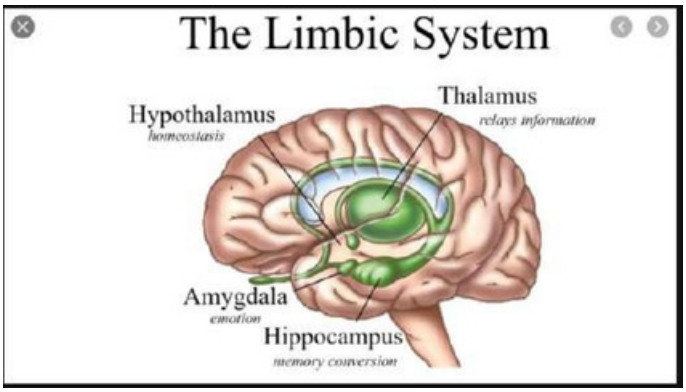

Figure 2 (Google Images)

Dopamine is synthesized in the ventral tegmental area (VTA) and is released into the NAc in response to rewarding stimuli. This release acts as a signal, reinforcing the behavior that led to the pleasurable experience (Blum, et al., 2022) (Febo, et al., 2017) (Della Torre, et al., 2018) (Gong, Fayette, Heinsbroek, & Ford, 2022) (Groman & Jentsch, 2012). In serious addiction, the pleasure syndrome cascades into the pain of the crash once the "high" effect of the drug wears off (Chung, 2018) (Nutt, Lingford-Hughes, Erritzoe, & Stokes, The dopamine theory of addiction: 40 years of highs and lows, 2015) (Rehm & Shield, 2019).

In the research gathered by the Grace Center for Healing, it was discovered when the amygdala, thalamus and hypothalamus points are electronically balanced with 30 hz frequency in auriculotherapy for 30+ days: relapse did not occur. The amygdala, thalamus and pituitary hormonal axis functions were achieving sustainable results short circuiting the fight/flight addiction loop (Frankiensztajn, Elliott, & Koren, 2020).

The involvement of other neurotransmitters cited in the Brain Reward Cascade were not tested. For instance, glutamate, the main excitatory neurotransmitter, was involved in synaptic plasticity, but was not tested (Febo, et al., 2017) (Della Torre, et al., 2018) (Gong, Fayette, Heinsbroek, & Ford, 2022) (Groman & Jentsch, 2012) (Tafet & Nemeroff, 2020) for the strengthening of neural connections in reward-related circuits (Blum, et al., 2022) (Febo, et al., 2017).

GABA, the main inhibitory neurotransmitter, plays a modulatory role in the regulation of reward-related behaviors; however, GABA production and opioid replacement therapy were not addressed due to contradictory chemical imbalances.

Research by Wise (2008) demonstrated that the Brain Reward Cascade is not limited to dopamine release in the NAc but involves a broader network of brain regions, including the prefrontal cortex and the amygdala. These regions contribute to decision-making, emotional processing, and the integration of reward-related information. Therapy was focused on these regions.

Although other studies have highlighted the significance of the Brain Reward Cascade in addiction, (Koob and Volkow (2010) emphasized the dis regulation of this Brain Reward Cascade to persistent drug-seeking behavior and loss of control in the brain cell death of the prefrontal lobe and executive functions of cognition. This need was addressed.

The repeated activation of the Cascade by addictive substances led to long-lasting neuroadaptations, altering the reward systems functioning and perpetuating addictive behaviors. The never-ending loop of the pleasure behavioral management in addiction (Roberto, Kirson, & Khom, 2021) (Gilpin, Herman, & Roberto, 2022) (Wolf, 2016) (Werner, Altshuler, Shaham, & Li, 2021) followed by the crash, the guilt and the shame, (Matthews, Dwyer, & Snoek, 2017) (Krentzman, 2013) (Coussens, et al., The Opioid Crisis and the Future of Addiction and Pain Therapeutics, 2019) accompanied the criminal outcomes.

The loop of HPA trauma destruction was addressed in the first order. The main focus of Wagner's research was to permanently disrupt the Cascade without drug intervention. (Frankenstajn, Elliott, & Koren, 2020). Secondly, how could the damaged epigenetic DNA codes be rewritten? What substances would restore the neuroplasticity of the damaged brain with DNA miswiring (Werner, Altshuler, Shaham, & Li, 2021) (Frankiensztajn, Elliott, & Koren, 2020) (Tafet & Nemeroff, 2020). Finally, the goal was to discover exactly which substances would restore neuroplasticity of the brain lobes to achieve brain coherence.

The restoration of brain coherence was of utmost importance. How could this be accomplished holistically keeping to the values of non-drugs care protocol? The clinical hypothesis of Dr. Wagner included the regulation of the fight/flight characteristics of the amygdala to the dopamine receptors.

If the HPA, amygdala axis (Frankiensztajn, Elliott, & Koren, 2020) could be short circuited, possibly the fight/flight response in the Brain Reward Cascade could be mitigated, then perhaps cravings would disappear. How could this be accomplished?

Moreover, recent studies have explored the role of the Brain Reward Cascade in non-substance-related addictive behaviors, such as gambling, internet and gaming disorders. (Volkow et al. (2017). These behaviors demonstrated similarities in the neurobiological mechanisms underlying both substance and behavioral addictions in genetic related disorders, suggesting a shared involvement of the reward cascade and the DNA, RNA genetic codes (Fuse-Nagase, Saito, Hirohara, & Miyakawa, 2015) (Hikitsuchi, et al., 2014) (Hikitsuchi, et al., 2014).

It was proven conclusively by ALL participants, that after the trauma issues were exposed and dealt with, 30 days of electro auricular therapy resulted in NO drug use. Reporting was done by one page written survey and YOUTUBE channel videos.

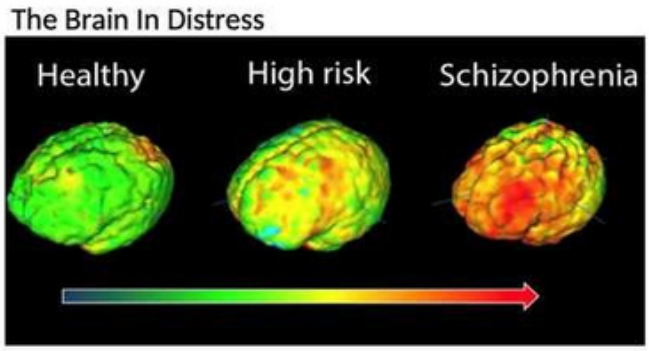

Figure 3 (Google Images)

Figure 3 illustrates images of the brain in distress. The brain on the left has a few “hot spots" (Boutet, et al., 2021) (Solo, et al., 2018) (Badgaiyan, et al., 2022) at the back of the skull indicating that perhaps there was a fall or bump on the head. This is common.

The second brain image has signs of serious distress and neuronal damage. This could be the brain of an athlete that had a baseball bat or a soccer shoe banged into his temporal lobe by a competitor. These brain cells are trying to achieve coherence and the cells are literally too hot to perform their functions. Notice the blue spots at the prefrontal lobe where the brain cells are cold. Here the traumatized brain already shows signs of death.

Finally, the far-right brain is the brain in serious distress because the prefrontal lobe has been seriously affected by drug use. The cells of the brain are so hot it cannot function. Frequently drug users will even notice the forehead is hot to the touch with no fever. The last image is a representation of a schizophrenia brain, but in actuality, it is the brain of a serious drug user. (Google Images Source).

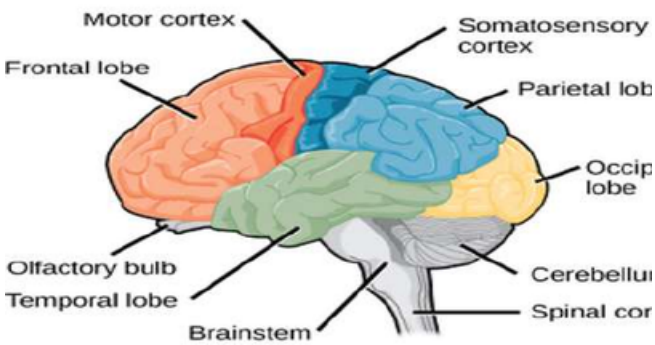

Figure 4.(Google Images)

Figure 4 indicates the proximity of the olfactory bulb where most illegal drugs are inhaled or snorted. The temporal lobe registering emotions of the individual are right below this lobe. The prefrontal lobe and motor cortex are severely affected by drug use inhaled or snorted through the nose. Finally, the somatosensory cortex and parietal lobe affecting movement are severely affected by drug use. The occipital lobe can be compromised by a fall or injury to the back of the head, or by using drugs such as benzodiazepines. It was noted in all cases that the cranial nerves affecting sight, hearing and the vagus nerve were damaged by drug use.

Prefrontal Cortex:

1. The prefrontal cortex, located in the frontal lobe, plays a crucial role in decision-making, impulse control, and judgment. Drugs such as cocaine, methamphetamine, and cannabis can affect this region. Prolonged drug use can lead to decreased prefrontal cortex activity, pollution and possible death of the cells with impairing cognitive functions reducing inhibitions. This results in impulsive behavior and poor decision-making, contributing to the cycle of incarceration and drug addiction.

Hippocampus:

2. The hippocampus, a region critical for memory formation and learning, is adversely affected by drugs like MDMA (ecstasy) and cocaine. These substances disrupt neurotransmitter systems, leading to memory impairment and difficulty in retaining new information. Long-term drug abuse can result in structural changes in the hippocampus, potentially leading to long-lasting cognitive deficits. Marijuana use can affect memory and learning processes. The hippocampus, a region crucial for memory formation, contains a high concentration of cannabinoid receptors. Activation of these receptors by marijuana can impair short-term memory and disrupt the encoding and retrieval of new information. Heavy and long-term marijuana use has been associated with persistent cognitive deficits, particularly in areas of attention, memory, and executive functions.

Basal Ganglia:

3. The basal ganglia are responsible for regulating movement, motivation, and reward. Located in the parietal lobe, stimulant drugs such as amphetamines and cocaine increase dopamine levels in this region, leading to intense feelings of pleasure and reinforcing drug-seeking behaviors. Over time, this can result in changes to the Brain Reward Cascade System and movement regulation.

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Amygdala:

4. The amygdala plays a central role in emotions and emotional memory. Drugs such as opioids, alcohol, and benzodiazepines affect this region, leading to an imbalance in emotional responses (Roberto, Kirson, & Khom, 2021) (Gilpin, Herman, & Roberto, 2022) (Wolf, 2016).

Excessive drug use can result in an overactive amygdala, leading to heightened anxiety, out of control aggression, and emotional instability. The amygdala is the center of flight/flight response. Unless the amygdala is regulated after cocaine use, these aggressive, impulsive behaviors continue.

Cerebellum :

5. The cerebellum, responsible for coordinating movement and balance, can also be impacted by drugs. The cerebellum responsible for coordinating movement and balance, can also be impacted by drugs (Werner, Altshuler, Shaham, & Li, 2021) (Frankiensztajn, Elliott, & Koren, 2020) (DeSabel, A, Kruse, Wolf, & Guenther, 2013) (Solo, et al., 2018).

Alcohol, for instance, affects the cerebellum, leading to impaired motor skills, slurred speech, and lack of coordination, inability to walk.

Ataxia is often an after condition of alcoholism. Other drugs, such as hallucinogens, can disrupt cerebellar function, leading to altered perception of space, time, and body movement

Hypothesis #3: The Effects of Neuropharmacology on the Brain Reward Cascade

● What are the effects of illegal drug use to the specific areas of the brain?

● Once cessation of drug use is achieved, while the Brain Reward Cascade is functioning, what nutrition rewires brain plasticity of the lobes?

● Can pure essential oils rewrite the A2 Allele DNA genetic codes?

● Does trauma of a country affect the type of illegal drugs manufactured in the exploitation
of citizens of that country?

Pharmacology of Drug Use in Japan

Cocaine	Coke, blow, powder, flake or crack	Sniffed, Smoked, Injected	Increased Alertness,excitation, increased Pulse, insomnia, mood swings, can get violent
Methamphetamines		Oral, injected, Smoked	Speeds up the mid to increase focus , and alertness, insomnia, used to work longer hours with more focus
Benzodiazapines		Oral, taken over the counter	Slurred Speech, Disorientation, Drunken Behaviour, Life

Table 1 (Google Images)

Illegal drugs have profound effects on various aspects of human health, especially the brain. The brain is a complex organ composed of different regions, each responsible for specific functions. When exposed to illegal drugs, these lobes or regions can undergo significant changes that disrupt normal brain functioning. Understanding the effects of illicit substances on specific brain areas sheds light on the dangers associated with drug abuse and addiction.

Effects of Illegal Drug Use on the Brain:

Cocaine: Snorted, sniffed, smoked, inhaled or injected with syringes. (Gardner, McGrath, Dowling, & Bai, 2022) (Manchikanti, et al., 2022) (Busserolles, et al.,

Replacement of current opioid drugs focusing on MORE- related strategies, 2020). Speeds up brain function, people use it to work harder, longer, increased pulse rate, insomnia, huge mood swings, schizophrenia after prolonged use (Shimazono & Matsushima, 1995) (Holder, Calaff, Maricque, & Tran, 2022) (Wagner & Bertram, 2020).

Methamphetamines : Oral, injected, smoked or inhaled through a pipe. Creates insomnia, is used by young professionals and executives with pressure of performance at their jobs. Used to work longer hours with increased alertness especially by young executives with pressures of performance over long durations (Holder, Calaff, Maricque, & Tran, Concentrated incarceration and the public-housing-to-prison pipeline in New York City neighborhoods, 2022).

enzodiazepines: Oral, over the counter drug causing slurred speech, slows down brain activity, given to children with ADHD tendencies, disorientation and drunk like behavior over a period of time (Marshall, Clinical trials.

Benzodiazepines: Oral, over the counter drug causing slurred speech, slows down brain activity, given to children with ADHD tendencies, disorientation and drunk like behavior over a period of time (Marshall, Clinical trials. Planned Ritalin trial for tots. A trial at this age heads into uncharted waters, 2000) (Pelham, et al., 1999) (Busby, Hatkevich, McGuire, & King, 2020) (Vargas- Medrano, et al., Psychological and neurobiological aspects of suicide in adolescents: Current outlooks, 2020).

Alcohol: Known as ethanol, alcohol strips the neuronal Mylene sheath (Miyamori, et al., 2022) (Antonelli, et al., 2021). Prolonged use of harsh alcohol can cause throat cancer or death by drowning in one's own blood when the arteries split.

Drug users use uppers to get high and wash down the drug with alcohol to come down creating complete havoc in the brain. Although alcohol is not considered an addiction in Japan or England; executives use it for closing business deals and bonding.

Marijuana: Legally grown in many countries as medical enhancement of drug relief of pain. Distorts perception and eyesight. Marijuana is dangerous to use while driving, affecting depth perception, distance calculations, perceptions of oncoming traffic.

Dappou herb in Japan is synthetic marijuana and Spice used in the US are also synthetic marijuana as (Fuse-Nagase, Saito, Hirohara, & Miyakawa, 2015) (Hikitsuchi, et al., 2014) (Taniguchi, Matsumoto, Kobayashi, & Wada,

2013) These synthetic drugs are so harmful to the brain, recovery is impossible to achieve. Although considered harmless and legal in several countries, marijuana is a very dangerous drug.

Dirty drug dealers use fentanyl on legal marijuana to increase the likelihood of addiction. Distortion of reality is common while using this drug.

Unsuspecting children die while using marijuana with a few grains of fentanyl. Fentanyl is 100 times more powerful than cocaine. This deadly substance cannot be seen, tasted, smelled or detected, and easily traps the unsuspecting. No amount of talk therapy can rewire the chemical decomposition of cellular brain functions.

iscussion #4 : The Global Trends Overview in Addiction Pharmacology-Pre/Post Covid

Before the Covid pandemic, over 59% of the world's children were on antidepressant medications (Chung, 2018) (Nutt, Lingford-Hughes, Erritzoe, & Stokes, The dopamine theory of addiction: 40 years of highs and lows, 2015) (Volkow & Blanco, 2021) (Chisholm-Burns, Spivey, Sherwin, Wheeler, & Hohmeier, 2019). Post pandemic, it is predicted by the global scientific community that ⅓ children are suffering with mental health, depression or suicidal tendencies. Canada is the fourth leader in written prescription drug use (Manchikanti, et al., 2022) (Busserolles, et al., Replacement of current opioid drugs focusing on MORE-related strategies, 2020). The US leads the world in antidepressant use with legal prescription use five times the entire global use (Pelham, et al., 1999) (Busby, Hatkevich, McGuire, & King, 2020) (Vargas-Medrano, et al., Psychological and neurobiological aspects of suicide in adolescents: Current outlooks, 2020).

These trends in global addiction reflect the anticipation of a coming pandemic of suicide in children (Volkow & Blanco, 2021) (Karila & Benyamina, 2019) (Matthews, Dwyer, & Snoek, 2017), incarceration in adults (Chamberlain, et al., Illicit substance use after release from prison among formerly incarcerated primary care patients: a cross-sectional study, 2019) and global increase in addictions (Manchikanti, et al., 2022) (Busserolles, et al., Replacement of current opioid drugs focusing on MOR-related strategies, 2020) (Coluzzi, et al., 2022) (Milroy & Parai, 2011) due to the trauma of the pandemic with lost loved ones, and the inability to socialize as a humanity.

Addiction Trends Specific to the Country of Japan:

"Japan has a problem with addiction; the number of addicted persons is estimated to be 100 000 for drug dependence (Milroy & Parai, 2011) (Miyamori, et al., 2022) 800 000 for alcohol dependence, 3.2 million for gambling disorders, and 4.2 million for adults with internet addiction (Pichini, Solimini, Beretta, Pacifici, & Busardò, 2018) (Hayashi, et al., 2021) (Chisholm-Burns, Spivey, Sherwin, Wheeler, & Hohmeier, 2019) (Gardner, McGrath, Dowling, & Bai, 2022) and adolescents with internet addictions (Kuss, Griffiths, Karila, & Billieux, 2014) (Jorgenson, Hsiao, & Yen, 2016) (Derevensky, Hayman, & Gilbeau, 2019) (Schou Andreassen, et al., 2016) (Kuss, Griffiths, Karila, & Billieux, 2014) (Cohen, 2009).Because there is a strong prejudice against addiction in Japan, sufferers do not readily admit they have addictions. Cultural shame destroys the families admitting to drug use.Family members are reported and taken to police (Karila & Benyamina, 2019) Therefore, actual cases may be much higher.

As leading experts in addiction research in Japan, members of the AS-SCJ 7 and the Japanese Medical Society of Alcohol and Addiction studies (JMSAAS) (Vargas-Medrano, et al., Psychological and neurobiological aspects of suicide in adolescents: Current outlooks, 2020) (Abdallah, et al., 2019) (Lewis, et al., 2021) (Lunghi, et al., 2022) (Cousins & Goodyer, 2015) (Hetrick, Merry, McKenzie, Sindahl, & Proctor, Selective serotonin reuptake inhibitors (SSRIs) for depressive disorders in children and adolescents, 2007). "8people participated in that study." (Milroy & Parai, 2011) (Miyamori, et al., 2022). Japan's clinical trials prove that patients engaging in SUD behaviour are chronically more disposed to committing suicide. Japan has one of the highest rates of suicide in the world (Volkow & Blanco, 2021), (Vargas-Medrano, et al., Psychological and neurobiological aspects of suicide in adolescents: Current outlooks, 2020). This thesis tracks the non-pharmaceutical methods of recidivism. Drug abuse in the country of Japan began after World War II and has been characterized by three epidemics of methamphetamine abuse (Holder, Calaff, Maricque, & Tran, Concentrated incarceration and the public-housing- to-prison pipeline in New York City neighborhoods, 2022) (Wagner & Bertram, 2020) followed by brain destruction with solvent abuse and the herb marijuana use of Dappou drug (Matthews, Dwyer, & Snoek, 2017), (Krentzman, 2013), (Coussens, et al., The Opioid Crisis and the Future of Addiction and Pain Therapeutics, 2019).

Since 1995, with the beginning of the third epidemic of methamphetamine abuse, drug abuse or dependence in Japan has changed markedly (Matthews, Dwyer, & Snoek, 2017), (Krentzman, 2013), (Holder, Calaff, Maricque, & Tran, Concentrated incarceration and the public-housing-to-prison pipeline in New York City neighborhoods, 2022). With a marked decrease in solvent abuse, the emergence of designer drugs such as "spice", and the introduction of the "dappou herb" has emerged (Coussens, et al., The Opioid Crisis and the Future of Addiction and Pain Therapeutics, 2019), (Fuse-Nagase, Saito, Hirohara, & Miyakawa, 2015), (Hikitsuchi, et al., 2014), (Tanibuchi, Matsumoto, Kobayashi, & Wada, 2013). Dappou herb (is a street drug typically produced by mixing herbs with synthetic cannabinoid (estimated to be the pharmacologically effective ingredient). This trend has recently spread to young people in Japan who consider the herb and cannabis to be THE new recreational drug (Fuse-Nagase, Saito, Hirohara, & Miyakawa, 2015), (Hikitsuchi, et al., 2014), (Tanibuchi, Matsumoto, Kobayashi, & Wada, 2013). As the drug is currently not legally regulated, the effects of addiction on Japan's youth are profound. 80% of all drug charges filed in Japan involve stimulants (Chamberlain, et al., Illicit substance use after release from prison among formerly incarcerated primary care patients: a cross-sectional study, 2019).

Use of this herbed cannabinoid can lead to sudden brain death, in the same way that a few grains of fentanyl (Fischer, Vojtila, & Rehm, 2018), (Ickowicz, et al., 2022) (Pichini, Solimini, Berretta, Pacifici, & Busardò, 2018), (Hayashi, et al., 2021) (Giorgetti, Centola, & Giorgetti, 2017), (IHan, et al., 2019). Legal Canadian marijuana can cause deaths among unsuspecting adolescents (Tanibuchi, Matsumoto, Kobayashi, & Wada, 2013), (Fischer, Vojtila, & Rehm, 2018), (Ickowicz, et al., 2022). The suicide rate of Japan's adolescent and young adult population has sky rocketed due to drug use. In other countries such as the USA and Canada, facts show the global incidents of suicide in children between the age of 10 and 16 is the third leading cause of death in children in 2023 (Vargas-Medrano, et al., Psychological and neurobiological aspects of suicide in adolescents: Current outlooks, 2020), (Lewis, et al., 2021). The USA leads the race in SSRI consumption, consuming five times the entire world consumption in ADHD medications with 11% of children addicted. Canada is in 4th place with 8% of the population addicted (Marshall, Clinical trials. Planned Ritalin trial for tots heads into uncharted waters, 2000), (Abdallah, et al., 2019), (Lunghi, et al., 2022). The legalization of marijuana in Canada has only led to an increase in fentanyl production by dirty dealers. This gets innocent children marijuana users ``hooked" in as little asa few grains of fentanyl or a few "tokes of weed" (Giorgetti, Centola, & Giorgetti, 2017), (Ickowicz, et al., 2022).

Commonly, it is said by psychiatrists, "Antidepressant drugs should not, however, be withheld for any reason, if indicated" (Cousins & Goodyer, 2015), (Hetrick, Merry, McKenzie, Sindahl, & Proctor, Selective serotonin reuptake inhibitors (SSRIs) for depressive disorders in children and adolescents, 2012). What will influence parents, psychiatrists and psychologists to seek natural remedies for the mental health conditions of today? (Findling, 2008), (Konstenius, et al., 2017), (Stolberg, 2002), (Marshall, Clinical trials. Planned Ritalin trial for tots heads into uncharted waters, 2000). Post 2007, Ritalin also become a major source of drug abuse affecting a large percentage of the population (Rehm & Shield, 2019), (Sprich, Safren, Finkelstein, Remmert, & Hammerness, 2016), (Konstenius, et al., 2017). Do not blame the adolescent brain subject to Ritalin use for ten years to crave cocaine. Opium, cannabis and stimulant drugs are prohibited in Japan. (Amphetamines, methamphetamines), including certain medicines for the treatment of ADD/ADHD (such as Adderall, Vyvanse and Dexedrine) (Stolberg, 2002), (Marshall, Clinical trials. Planned Ritalin trial for tots heads into uncharted waters, 2000), (Pelham, et al., 1999). These stimulants SSRIs widely prescribed in other countries are strictly prohibited and illegal to bring into Japan (Hayashi, et al., 2021), (Chisholm-Burns, Spivey, Sherwin, Wheeler, & Hohmeier, 2019) (Gardner, McGrath, Dowling, & Bai, 2022).

Predominant trends in Addiction in Japan since WWII:

● The emergence of non-regulatory drugs represented by designer drugs such as “dappou herb”. These imply the change from "hard drugs" to "soft drugs”, are criminally unregulated and very dangerous (Coussens, et al., The Opioid Crisis and the Future of Addiction and Pain Therapeutics, 2019), (Fuse-Nagase, Saito, Hirohara, & Miyakawa, 2015).

● Technology addictions like gaming and internet disorders consume 10-15% of the Japanese adolescent population (Miyamori, et al., 2022), (Kuss, Griffiths, Karila, & Billieux, 2014), (Derevensky, Hayman, & Gilbeau, 2019).

● Alcohol binge drinking in workaholic stressful corporations is the third most prevalent addiction (Coluzzi, et al., 2022), (Milroy & Parai, 2011), (Miyamori, et al., 2022). . It is not uncommon for drunks to be seen sleeping in the streets of Japan. Most conventional doctors cannot successfully treat addiction because they are inadequately trained and tend to prescribe mood altering drugs, replacing one addictive substance with another, leaving addicts at high risk of death and relapse (Coluzzi, et al., 2022), (Konstenius, et al., 2017), (Van Emmerik-van Oortmerssen, et al., 2014).

● Adolescents are using over the counter, SSRI’S, prescription drugs legally and dying as a result. 64% of the population uses non-prescription drugs such as analgesics, tranquilizers, and sleeping pills such as benzodiazepines (Pelham, et al., 1999), (Busby, Hatkevich, McGuire, & King, 2020), (Vargas-Medrano, et al., Psychological and neurobiological aspects of suicide in adolescents: Current outlooks, 2020). Benzodiazepines are the second most prevalent drug after Meth addiction (Becker & Correll, 2020), (Shimazono & Matsushima, 1995). Benzodiazepines can be deadly used for an extended period (Van Emmerik-van Oortmerssen, et al., 2014), (Marshall, Clinical trials. Planned Ritalin trial for tots heads into uncharted waters, 2000), (Pelham, et al., 1999), (Busby, Hatkevich, McGuire, & King, 2020). These drugs consistently led to cocaine, heroin and methamphetamines use.

Other factors:

● Cannabis use is three times more prevalent than Meth with herbal enhancements called dappou drug. Because dappou is herbal, it is not considered illegal. Synthetic cannabis is lethal to the brain (Shimazono & Matsushima, 1995), (Coussens, et al., The Opioid Crisis and the Future of Addiction and Pain Therapeutics, 2019).

● MNDA is distributed in nightclubs as an ecstasy drug promoting sex drive.

● In Japan, people are publicly drunk, going to work “hungover”. Culturally, public drunkenness is tolerated, considered an integral proof of the work culture in Japan. Both beer and sake are available in vending machines . (Milroy & Parai, 2011), (Miyamori, et al., 2022).

● Over 1,000,000 Japanese are diagnosed with alcohol disorder, only 80,000 are seeking treatment (Milroy & Parai, 2011), (Miyamori, et al., 2022).

● 2.6 Million in Japan are drug users. Heroin use is a crime, meth and cocaine are gaining popularity as alternative drugs (Holder, Calaff, Maricque, and Tran,Concentrated incarceration and the public-housing-to- prison pipeline in New York City neighborhoods, 2022), (Wagner & Bertram, 2020).

● Only 14,755 in 2018 were arrested for use of illegal drugs (Blum, et al., 2022), (Febo, et al., 2017).

● Police focus on Methamphetamine use and crime with 80% of all arrests originating with this drug (Chamberlain, et al., Illicit substance use after release from prison among formerly incarcerated primary care patients: a cross-sectional study, 2019).

“During World War II, meth use was encouraged by the Japanese state to enhance the productivity of factory workers and to in-still national duty to fight in battles,” Brewster said. “Following the Japanese defeat, the use of the drug spiraled into an epidemic. The act of choosing illegal drugs was deemed to be innately selfish, socially and morally polluting.” (Karila & Benyamina, 2019), (Coussens, et al., The Opioid Crisis and the Future of Addiction and Pain Therapeutics, 2019), (Fuse-Nagase, Saito, Hirohara, & Miyakawa, 2015), (Hikitsuchi, et al., 2014).

Miriam Kingsberg, a specialist in Japanese history at the University of Colorado, (2013) that this wave of meth addiction “came to symbolize a collective identity of defeat, despair and dependence to truly regain self-sovereignty, Japan had to be drug-free.” Japan’s postwar meth epidemic, and two subsequent waves of meth addiction linked to economic downturns in the 1970s and 1990s, led to tough sentences for first-time meth offenders and the creation of specialized drug enforcement officers. “Yet Japan’s zero-tolerance attitude to drugs comes at a cost. Because using drugs has such a big stigma attached to it (Karila & Benyamina, 2019).It is harder for people to admit they have a problem and to get help when a family can jail an addicted family member instead of seeking rehabilitation.

The less people who seek help, the less treatment the government should provide and the smaller the number of registered addicted users. These users are never tracked and their rehabilitations are not registered in such systems as blockchain.

This stigma is also reflected in Japan's almost non-existent drug death statistics, where overdoses are politely labelled under 'heart failure' and where autopsies performed in a small proportion of suspicious or suicide deaths (Karila & Benyamina, 2019), (Gardner McGrath, Dowling, & Bai, 2022), (Manchikanti, et al., 2022), (Busserolles, et al., Replacement of current opioid drugs focusing on MOR-related strategies, 2020). Since 2014 and the emergence of designer drugs such as "spice" with the introduction of the "dappou herb " has emerged. Dappou herb is a street drug typically produced by mixing herbs with synthetic cannabinoid (estimated to be the pharmacologically effective ingredient) (Fuse-Nagase, Saito, Hirohara, & Miyakawa, 2015), (Hikitsuchi, et al., 2014), (Tanibuchi, Matsumoto, Kobayashi, & Wada, 2013). This trend has recently spread to young people in Japan who consider the herb and cannabis to be THE new recreational drug. As the drug is currently not legally regulated, the effects of addiction on Japan's youth are profound. 80% of all drug charges filed in Japan involve stimulants. ** NOTE: The damage to the brain in use of dappou cannabinoid is not reversible, and can lead to brain death (Fuse-Nagase, Saito, Hirohara, & Miyakawa, 2015), (Hikitsuchi, et al., 2014), (Tanibuchi, Matsumoto, Kobayashi, & Wada, 2013).

Clinical Evaluation and Clinical Trial Conclusion :

Electrical stimulation of the HPA axis using auricular acupuncture on specific limbic/parasympathetic HPA axis neuronal points, combined with near red and infra res light therapy were administered to heal the prefrontal, parietal, somatosensory and occipital lobes. The results were reported by oral findings and a written survey. These measures directly reduced mood depression, simultaneously increasing agility and speed in movement. These were not double blind studies. In lieu of drug inhalation, the participants chose pure essential oils replacing inhalation of meth, cocaine and heroin.. The inhalation of specific essential oils were later tailored to the needs of each individual participant. These natural plant therapies seemed to ease their stress, brain disturbances, psychosis, as well as being enjoyed by the "inhalation" patients. Pure monoterpene oils such as Angelica, Peppermint, Sacred Frankincense, Spruce, Northern Lights Black Spruce, Cypress , and Thieves oil were inhaled to potentially rewrite the miswritten information coded into cellular memory caused by the neuropharmacology of drug use.

DNA encoding tests in future double blind trials would be required to see if genetic changes could be achieved. Sesquiterpene oils, such as Vetiver, Sandalwood, Patchouli, and Myrrh may have broken down the generational DNA coding to end generational coding. There is evidence in Dr. Kenneth Blum's work showing that DNA generational coding in addiction cases may be a critical point in achieving recidivism. This research would require governmental funding.

Angelika Wagner's research was to try to prove recidivism is achievable without pharmacology and drug use. Sesquiterpene oils were used for inhalation. These essential oils seemed to be a powerful remedy replacing inhalation tendencies post eradication of inhalation drugs. In over 100 cases of holistic therapy at Grace Center Healing essential oils have proven effective, efficient, long lasting promoting human well being.

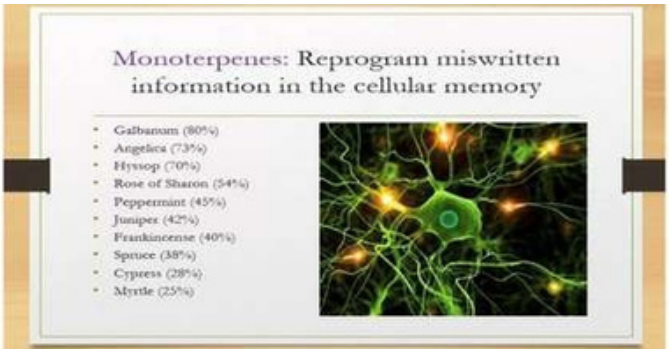
The blends of Purification or Valour oil were incorporated to boost mood and motivation. The hertz frequencies of pure essential oils ranges from 166 HZ to 660 HZ. In all drug use addiction cases, heart damage is evidenced as a comorbidity.

The study of the chemical effects of essential oils and plant therapies is a separate lifetime education. It is the study of the interactions of specific plants with the brain and body chemistry.

Research cited is research done by the founder and scientists at Young Living Company who have been pioneers in the field of phyto chemistry for over 30 years. Participants complained of fatigue, lack of motivation and inability to do physical exercise or work, prior to the trial. Post trial, none of these behaviors were exhibited. A double blind study with changes and verification of the genetic codes pre and post trial is recommended. There was no effort or damning exercise of "will". Once the auriculotherapy was applied, cravings ceased.

Once near red and infra red light therapy, amino acids and essential oils replaced the toxic brain chemistry, each individual reported they had no interest in continuing any drug habits. This was still not enough to regulate the body systems. Serious heart damage was noted in each case, sensory organ damage and parietal lobe damage were also noted post trial.

The general public is unaware of the damage caused by drug use to the brain and organs. It is an archaic and appalling theory that a person has a moral failing due to a hijacked brain caused by an early childhood trauma with subsequent prescription drug use legally prescribed to a child or young adult in order to escape that pain of the trauma



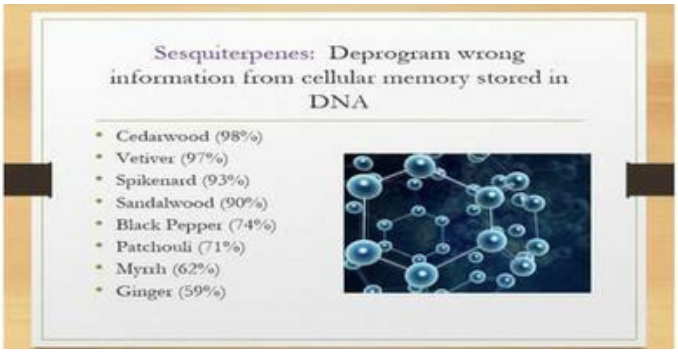


Figure 5. (Essential Oils Integrative Medical Guide. D. Gary Young, Founder of Young Living)

The hypothesis of the pharmacology effects changing the firing of the nerve impulses to the specific lobes was addressed with the following sequence.

1) The thalamic addiction response, the serotonin/dopamine response, was addressed with 30 hertz electronic frequencies on the auriculo-therapy points, using a device called the Stim Pro plus device (Febo, et al., 2017), (Della Torre, et al., 2018). Dr. Holder had previously discovered the parasympathetic points and the limbic points. Dr. Angelika Wagner discovered the amygdala, hypothalamus and pituitary axis points. (HPA axis) (Roberto, Kirson, & Khom, 2021), (Gilpin, Herman, & Roberto, 2022), (Wolf, 2016), (Werner, Altshuler, Shaham, & Li, 2021), (Tafet & Nemeroff, 2020). Wagner also discovered healing of the cranial nerve points, 3,4,6 for eyes and 5, 9,10 for vagus nerve tone regulation. Balancing of these points by an electric device (Stim Pro PLUS) were measured by the auriculotherapy until the device showed readings of zero on three consecutive days.

2) Specific points addressed distress in the cranial nerves leading to sensory organs. The findings revealed short - circuiting the relay function of the chemical imbalance of fight /flight amygdala response, the thalamic addiction response, the serotonin/dopamine response (Della Torre, et al., 2018), (Gong, Fayette, Heinsbroek, & Ford, 2022. Illicit substance use after release from prison among formerly incarcerated primary care patients: a cross-sectional study, 2019), (Roberto, Kirson, & Khom, 2021), (Gilpin, Herman, & Roberto, 2022), (Frankiensztajn, Elliott, & Koren, 2020), (Becker & Correll, 2020) caused by drug medication, the prefrontal lobe executive function was eventually restored to working capacity. This was measured by the specific points being consecutively stimulated, cravings for drug use ceased after 5 days, Improvement in eyesight, digestion and movement was verbally reported after 30 days. Within 90 days, each person reported ability to work, and drug addiction ceased with complete permanent withdrawal occurring. This was not a double blind trial. Further pre and post testing on sensory organs would require a different team of experts' verification .We considered the results to be conclusive due to participants' verbal and written report.

3) Based on claims by Kenneth Blum, PhD that rewriting the genetic codes was required for permanent recidivism, these theories require a separate genetic team of scientists and further clinical trials. Although changes in the genetic codes may have been accomplished with the use of pure distilled plant therapies and essential oils, this is medically inconclusive without genetic code testing.

It was noted that pure essential oils were offered for inhalation and topical application pleasurable and soothing responses in the moods of participants. Also noted, but not proven : the olfactory lobes and pre-frontal lobes providing support to the executive function and decision making may also have been healed. These methods support emotional childhood trauma as the root cause of the addiction (Frankiensztajn, Elliott, & Koren, 2020), (Tafet & Nemeroff, 2020), (Boutet, et al., 2021). Again, further testing using fMRI imaging scans of the prefrontal lobe are required for conclusive brain evaluation. The sequences of therapies remained the same in both groups who remained in clinical trials for 90 days or longer. The outcomes in both groups were identical. No relapse addiction from 2013-2023, no suicidal ideation from 2017-2023. All established protocols were tailored to meet each individual's needs.

Summary Results of Three Clinical Trials:

● 95 people participated, 60 people who participated in treatment did not stay the required 90 days to be included in results of the trial (Group 1).

● 33/35 people incarcerated, convicted long term detained inmates have not relapsed since 2013. All participants contribute to society through honest legal work and university education (Group 2).

● 6/6 suicide adolescent girls have not relapsed into suicidal ideation since 2017. All participants graduated high school and have not resumed prescription drug use of any kind. (Group 3). These subjects' results have not been documented since 2021. All participants in the three trials exhibited social and occupational impairments, mental and psychological debilitation due to persistent drug use with comorbidity of brain and organ dysfunctions prior to the clinical trial. These conditions were self reported verbally and in written summary by each participant. All participants experienced social, emotional, chemical or physical childhood trauma leading to their accumulated addictions. This was not a double blind study. This was a privately funded trial of the Grace Center for Healing over a period of 10 years. Human subjects in accordance with the guidelines of the USA Federal guidelines for clinical trial protocols volunteered for this study. All regulations were followed under the protection of the Federal Guidelines of human subjects in the USA. The regulations found at 45 CFR part 46 are based in large part on the Belmont Report and were written to offer basic protections to human subjects involved in both biomedical and behavioral research conducted or supported by HHS. Participants in the LA clinical trial exhibited mental debilitation with multiple drug addictions, years of isolation and incarceration, movement disorders, inability to read well, hear well or see well. Their debilitations included mental health issues of suicidal ideation and deep depression. Of these 95 participants, 33/35 participants who remained in the trial for 90 days and have remained permanently sober since 2013.

Trial Number three was conducted without regulation in private practice in Toronto, Canada (2017-2021) 6 adolescent girls exhibiting suicidal ideation due to prescription drug use participated. They have not relapsed in 6 years. These The adolescent study was conducted in a different country, producing parallel protocols with parallel results. This situation must be noted. Given the current statistics of a possible suicidal pandemic in children and adolescent due to the volume of SSRI use in USA, and other countries, it is great importance that public health authorities be made aware of increased prescription drug use in treatment of children's mental health. A tsunami of adolescent and young adult deaths can be anticipated due to the escalation of early drug use, with combinations of cyber bullying, social media and internet addiction. Clinical Observations Movement, sensory information seemed to be restored by 30 H. cranial nerve stimulation, but the amount of rehabilitation was not measured, only reported.Biophotomodulation created by Vielight in Toronto seemed to enhance functioning of the prefrontal lobe, the parietal lobe, the occipital lobe and accompanying depression.

. Inhaling essential oils served as substitution for inhalation of the illegal drugs and decoding of the genetic information (Frankiensztajn, Elliott, & Koren, 2020), (Tafet & Nemeroff, 2020). Without MRI scans, oral reports and visual documentation indicated neuroplasticity of lobes' restoration. Without live blood genetic analysis, it cannot be determined if or by whom much the genetic codes of the A2Allele were changed with these processes. All participants reported increased hearing, eyesight, mobility and cognitive functions and were able to return to work or go to university to complete studies. Families reported an improved ability to perform social, interpersonal, and occupational functions. Evaluations were done with individual written one-page surveys. Psychological testing, questionnaire on executive and sensory function were completed pre-trial.

Testimonial videos and written surveys were completed post trial. Post trial evaluations: only the incarcerated population was treated as a group with video and written surveys. The suicidal ideation trial of the adolescent girls was done individually, no cross- collaboration with the participants. No videos of the girls were taken due to the sensitivity of the information, cyber bullying and social media attacks. The participants did not know each other. Each trial completed 90 days or longer with each individual's needs tailored to their specific evaluations, protocols and brain nutrition. All participants reported the ability to return to work and education. They were no longer a burden on society or their families. This open-protocol data suggests that placebo-controlled studies are warranted for further study and research. This was not a double blind study.

Program Evaluation, Program Trial Evaluation:

Future Study Recommendations : Environments of Hospital or University

: fMRI brain scans,

: Live blood cell diagnostics,

: Vision and hearing testing

: Biomedical team before and after the trials

: Complete medical examinations might include oxi-meter readings in order to measure oxygen capability and exercise to tolerate heart blood flow capacity. (Not funded)

: Epigenetic coding of live blood cell culture was not tested, therefore changes in genetic codes remain undetermined.(Not funded)

: Heart evaluations, mitral valve apex readings and right valve pressure to be taken pre and post trials.

Changes in epi-genetic blood codes were not studied or determined to further complete the findings of Kennth Blum, PhD.(Werner, Altshuler, Shaham, & Li, 2021) (Frankenstajn, Elliott, & Koren, 2020) (Tafet & Nemeroff, 2020) (DeSabel, A, Kruse, Wolf, & Guenther, 2013) (Boutet, et al., 2021). Further neurological readings of spinal column tone (Dr. Jay Holder) were also not taken. To be completely thorough in all aspects of generational recidivism in addiction and to acknowledge the decades of studies already completed by other scientists, it is recommended that changes in the epigenetic codes require a different scope of expertise and a different trial. It is also recommended fMRI brain scans accompnay all findings to view the articulation of the brain as changes in behcaiour and brain chemistry occur. The addiction cycle of over 40 people was holistically addressed in this self-funded trial spanning a decade of clinical determination. This trial was a global first and Guinness Book of World Record Accomplishment.

The Rapid Brain Therapy Method for Addiction was discovered by Angelika Wagner. Global Guinness Book World Record, July, 2023.

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